

GeoYuan

Project Development Timeline

2026.03.30 — 2026.07.14

Phase 1: GyID Foundation

2026.03.30 — 2026.04.12

- 03.30 Project created — Rust Workspace initialized
- 04.01 Toolchain: stable-gnu, MSYS2, GCC 15.2.0
- 04.04 Offline mode + HTTP timeout + Validator fix
- 04.05 Chain anchoring: Polygon + Aptos; GUI v0.3.1
- 04.07 Critical fixes: Default trait, LocalStorage, C FFI
- 04.08 Geoyuan website (Astro 4, cyberpunk, 13 pages)
- 04.12 Reverse geocoding + IP multi-source failover

Phase 2: Distributed System

2026.04.14 — 2026.04.29

- 04.14 geoyuan-core: identity, crypto, geo, photo, wallet
- 04.23 P2P module: libp2p Swarm, mDNS, Kademlia
- 04.29 geoyuan-core: 94 tests all passed
- 04.29 geoyuan-cli: created from scratch
- 04.29 geoyuan-gui: full features, zero warnings
- 04.29 Wallet sync + Multi-device linking

Phase 3: GPv1 Protocol [Core Innovation]

2026.05.07 — 2026.05.09

- 05.09 Ed25519 real signature verification
- 05.09 H3 GeoCast geographic broadcast (Res 12, ~9m)
- 05.09 PoL Proof of Location (GPS + Ed25519 signature)
- 05.09 CLI p2p: start / status / peers / peers-nearby
- 05.09 GUI achievement & energy system integrated
- 05.09 107 tests all passed (85+21+1)
- 05.09 P2P deployment scripts + Operations guide

Phase 4: Documentation & Planning

2026.07.14

- 07.14 GPv1 Protocol Whitepaper (16 chapters)
- 07.14 Development Status overview (6 done, 10 pending)
- 07.14 Competitive Analysis (11 international projects)